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Serving Adults Adolescents and Children

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Neuropsychological Evaluation
Privileged and Confidential

Patient: David Nimer

DOB: 05/31/1991

Age: 26-11

Date of Primary Injury: 02/07/2015

Dates Tested: 05/16,18,19/2018

Date of Report: 07/04/2018

Background and Referral Information: Mr. David Nimer was referred for a neuropsychological re-evaluation to assess his recovery following the onset of a number of physical, cognitive and emotional difficulties resulting from a single-vehicle motor vehicle accident in Oklahoma on February 7, 2015. A current follow-up neuropsychological evaluation is reasonable and necessary to establish his present neuropsychological profile and level of functioning, identify what neuropsychological deficits and assets may be present, and determine the status of his recovery. The evaluation will also indicate whether further treatment, educational and/or rehabilitation interventions would be recommended.

REVIEW OF BACKGROUND:

Motor Vehicle Accident History with Review of Available Records:

Mr. Nimer reported that he has been in four motor vehicle accidents (MVA); records indicate that the accident of record resulted in a head injury and other physical trauma. Records were not available for the other auto accidents, so this examiner is documenting the patient's report of those events. The first occurred when he was 19; he was a restrained front-seat passenger in a roll-over accident. When he was 22 he was the driver in the second MVA, another roll-over accident. In both instances he had no loss or alteration of consciousness or pre- or post-traumatic amnesia, and did not sustain any other serious physical injury.

Mr. Nimer was the driver in the 2015 accident of record, when the vehicle struck a tree at an estimated 60 mph. This examiner has reviewed considerable medical records provided by Mr. Nimer; records available for review do not include Police Report or EMS records. After hitting the tree the vehicle caught fire, and Mr. Nimer was trapped in it for up to a few minutes while he was unconscious. A passerby, a driver who had been driving behind him, pulled from the vehicle. Mr. Nimer regained consciousness at some point after being pulled from the vehicle. He also reported anterograde amnesia of about an hour prior to the crash. When EMS arrived on the scene they found him to be minimally responsive; he was promptly intubated, as there was concern about possible airway burn.

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Mr. Nimer was brought by EMS to OU Medical Center, where he was admitted to the trauma unit, diagnosed with a Traumatic Brain Injury (TBI) and given an initial Glasgow Coma Scale (GCS) rating of 3. He was put in an induced coma, a procedure typically intended to reduce swelling in the brain following TBI. In addition to that injury, OUMC records document injuries to his face, arm, hip, thigh and leg; in particular, there was an open proximal fracture of the right humerus which required immediate surgery.

Mr. Nimer underwent CT and serial MRI scans of the brain over the next 48 hours. Initial radiology reports indicate some bilateral involvement of the frontal lobes, as well as a very small subdural hematoma overlaying the occipital lobes. However, there was no midline shift, no mass effect and no hydrocephalus, and sinusoidal cavities were within normal limits. Other records indicate that four days later, having had the induced coma terminated, Mr. Nimer had begun to recover some basic cognitive functions, as seen in improved orientation and sequencing. He was discharged to Valir Rehabilitation on the fifth day of his hospitalization at OUMC. A Discharge Summary reported that Postconcussive Syndrome (PCS) accounted for the difficulty following directions and impulsivity (which had been addressed with restraints) during what was described as an otherwise unremarkable hospital course.

Mr. Nimer told this examiner that he stayed at Valir Rehabilitation only a few days before being discharged to his family's care. He was followed by neurology and had an additional MRI on March 2015. The Radiology report found indications of bilateral axonal micro-shearing which appeared to be stable by that time; this confirmed that a TBI had occurred (i.e., "evidence for TBI persists".) The last Radiology report provided to this examiner documents a CT scan in June of 2016. It states that there were some very small areas of unusual signal. These were judged most likely to be "artifacts" (not indicating abnormality).

Last, Mr. Nimer reported a fourth car crash had occurred in October 2016, about 17 months after the accident of record. He hit the back of the car ahead of him at a speed of about 35 mph, by his report. The airbag deployed and he reported being "stunned", but had no loss of consciousness; his report of being "stunned" is a neurological sign consistent with concussion.

Social/ Developmental History:

Mr. Nimer completed an extensive Social/Academic/Medical History form which is a routine part of this evaluation. He consulted with his mother to obtain information about her pregnancy with him, his family medical history and his early developmental milestones. He is one of three sons born in an intact family system. There is some history of psychiatric issues among relatives on his mother's side of the family, and of some occurrence of Attention Deficit Hyperactivity Disorder (ADHD) on his father's side; three grandparents have Alzheimer's Disease. His father earned a Juris Doctorate and practices law. His mother earned a Master's degree in Business Administration (M.B.A.), and pursued a career in the renewable energy field.

Mr. Nimer was the result of his mother's first pregnancy, one of a pair of twins. This resulted in more-than usual medical supervision of the pregnancy, by family report. She was given medication to mitigate contractions that began at 31 weeks. She incurred toxemia in her 38th week, leading to delivery of the twins by C-section. His weight was 6 lbs 11 oz. While APGAR scores are not available, the patient cried immediately upon delivery, and there were no reported complications following the birth such as jaundice, respiratory problems, etc.

His mother cannot recall his age at the time of at the emergence of his developmental milestones, as she was dealing with lack of sleep after the birth of twins; however, she was able to report that their emergence gave her no cause for concern, based on her experience with her three children. David had none of the behavioral problems some young children experience, such as bed-wetting, sleepwalking, head-banging, rocking, nail-biting, or temper tantrums. He had no episode of high fever during his childhood, and made steady developmental progress without any period in which he regressed or lost skills.

Academic History:

Documentation such as report cards was not available for review by this examiner, but Mr. Nimer apparently had no difficulty learning basic academic skills in the early grades, by his mother's report. He attended private school through high school, graduating in 2010. He first enrolled at Dickenson College, where he began to struggle with academics. He transferred to the University of Colorado-Boulder, where he completed a Bachelor of Arts degree with a reported GPA of 3.3. He took the Graduate Record Exam (GRE) standardized test twice in the spring of 2018, achieving somewhat stronger scores on the second administration. At that time he achieved scores at the 97th percentile on Verbal Reasoning, at the 87th percentile on Quantitative reasoning and at the 93rd percentile on Analytical Writing.

Additional Relevant History:

While at Dickenson College, Mr. Nimer was evaluated by Licensed Educational Psychologist Lisa Maheras, ?? [degree]. His younger brother had been diagnosed with ADHD, and he and his parents wondered whether undiagnosed ADHD might be a reason for his academic struggles. It was a relatively thorough evaluation that included tests of intelligence, memory, achievement and ADHD rating scales. Neither problems with attention nor hyperactivity were found to be at a clinically significant level in that evaluation.

After transferring to CU, he and his parents continued to be concerned about his academic achievement. In April of 2014 psychiatrist Dr. Brian Hersch reached a provisional diagnosis of ADHD and prescribed Vyvance and Adderall intended "to help Mr. Nimer finish that semester", with the stipulation that he pursue formal testing.

Mr. Nimer initiated a second ADHD evaluation through the campus' Wardenberg Health Center. In the report of that evaluation, Mr. Nimer's parents reported that he had been quite impulsive from an early age, a trait they felt has continued into adulthood. His parent said this had occasionally resulted in untoward consequences. That report also noted their concerns about longstanding problems with procrastination and with time management.

That evaluation for ADHD included Barkley's ADHD Rating Scale and an ADHD continuous performance test (Integrated Auditory and Visual CPT [IVA Plus]), as well as screening for depression and anxiety (Beck scales) and a personality inventory (MMPI-2). The Validity Scales on the MMPI-2 indicated that he had responded accurately and consistently. There apparently were no indications of any attempt to either exaggerate or minimize problems, and it was considered a valid and interpretable profile. The "slight [clinical] elevations on Scales 4, 7 and 9)" were consistent with his self-report and that of his parents in regard to impulsivity, excess energy and difficulty adhering to usual social standards and expectations. The examiner added that he was likely to feel tense worried and possibly agitated at times, and was further considered likely to have a great deal of difficulty concentrating. (Note that this psychological contribution

to this difficulty is not the same as the cognitively-driven deficits generally considered to be at the root of ADHD as it is usually understood). His Beck responses were consistent with mild depression, although the more thorough exploration of depression on the MMPI was not clinically significant.

His response style on the IVA Plus standardized test suggest that he put forth good effort and motivation. Basic reaction times were in the above-average to superior ranges (presumably for his age group). He had significant problems with auditory attention and with focus, especially under low-demand or low-stimulus conditions; these problems fell in the severely impaired range. These were not seen when the visual stimuli were presented. However, his difficulty with distractibility was much more troublesome for visual information and distractions than it was in the auditory realm. The examiner determined a diagnosis of ADHD, Combined Type.

In February 2016, Mr. Nimer pursued a neuropsychological evaluation at The Menninger Clinic. A neuropsychological evaluation is a more thorough assessment than the previous evaluations he had obtained, and more suited to assessing the effects of TBI on cognitive functioning. Using the data from the 2011 evaluation for comparison, that examiner found no areas in which cognitive skills had declined, in spite of the intervening head injuries, and several areas which showed significant improvement. Therefore the examiner said it was unlikely that Mr. Nimer had “significant persistent cognitive sequelae associated with TBI”. Mr. Nimer made the data from that evaluation available; while that evaluation did not have the scope of the present assessment, where possible that data was compared with the current data to better assess Mr. Nimer’s recovery and his current neuropsychological status.

At the time of the 2016 evaluation the patient was given the Rivermead Post-Concussion Symptoms Questionnaire, a self-report measure of symptoms associated with PCS, and that was re-administered here. In the earlier assessment he had endorsed significant difficulties including fatigue and memory problems. At this time he did not cite any of those areas as troublesome or problematic.

Diagnostic Criteria for Grade of Concussions, Traumatic Brain Injury (TBI) and Postconcussive Disorder (PCS):

It is documented that the patient sustained a front impact blow in February 2015 due to his vehicle hitting the tree, and experienced neurological symptoms including anterograde amnesia and altered consciousness. The sudden and complete stop of the vehicle at that time represents a whiplash (or coup - contra-coup) injury. Such scenarios typically result in a relatively more serious injury, since acceleration-deceleration forces occurring in the head can be 2 to 2½ times the forces occurring in the vehicles.

Medical guidelines regarding severity of head injuries indicate that concussions are ranked in three Grades or levels, based on symptoms at the time of a blow to the head: Grade 1, in which the individual was confused or had altered consciousness, but remained conscious, and symptoms cleared within 15 minutes; Grade 2, similar to Grade 1 but with symptom duration exceeding 15 minutes; and Grade 3, in which there is loss of consciousness for a few seconds or more, followed by notable disruption of cognitive functioning.

The available data indicate that the patient sustained a Grade 3 concussion in the February 2015 accident of record. A widely accepted definition of Mild Traumatic Brain Injury (MTBI) has been established by the American Congress of Rehabilitation Medicine (ACRM, 1993) and names four inclusion criteria; only one of the following is necessary to support a diagnosis of TBI.

1. Any period of loss of consciousness;
2. any loss of memory for events immediately before or after the accident;
3. any alteration of mental state at the time of accident (e.g., feeling dazed, disoriented, or confused);
4. focal neurological deficit(s) that may or may not be transient.

(Note that loss of consciousness is not necessary to meet medical criteria for Traumatic Brain Injury [TBI]).

A medically-induced coma following the accident of record adds a level of complication to the determination of the level of severity of his head injury. It may best be termed a Mild TBI – Complicated.

Post Concussive Syndrome consists of a set of symptoms that can develop after TBI. Most individuals who sustain Mild TBI will demonstrate only a few symptoms of Post Concussion Syndrome (PCS). In most cases, these typically resolve within three months; however, these symptoms persist in about 15% of MTBI patients. The patient's report on the Rivermead Post Concussion Symptoms Questionnaire indicates that he is not aware of any PCS symptoms at this time.

TESTS ADMINISTERED:

Wechsler Adult Intelligence Scale – Third Edition (WAIS-III)

Academic Achievement Battery (AAB)

Adult Basic Learning Examination (ABLE, Reading Comprehension Subtest)

Wechsler Memory Scale-Revised (WMS-III)

(Paired Associate Learning, Logical Memory and Visual Reproduction Subtests, immediate and delayed presentations)

Rey 15 Item Test

Rey Verbal Recognition Test

Rey Auditory Verbal Learning Test (RAVLT)

Rey Complex Figure Test (ROCF)

Rey Tangled Lines

Word Associations Subtest – Animals

FAS Verbal Fluency Subtest

Grooved Pegboard Test

Hooper Visual Organization Test (VOT)

Beery Visual-Motor Integration Test (VMI)

Stroop Neuropsychological Screening Test

Test of Variables of Attention (TOVA)

Test of Memory Malingering (TOMM)

Halstead-Reitan Neuropsychological Test Battery for Adults (15 and older)

- Speech Sounds Perception Test
- Rhythm Test
- Trails A

- Trails B
- Sensory-Perceptual Examination
- Tactile Form Recognition Test
- Lateral Dominance Examination
- Grip Strength
- Aphasia Screening Test
- Manual Finger Tapping Test
- Tactual Performance Test
- Category Test

Millon Clinical Multi-Axial Inventory (MCMI-III)

Rivermead Post Concussion Symptoms Questionnaire

Adult Self-Report Scale (ASRS – Screening Checklist for Adult ADHD)

Mood Disorder Questionnaire (MDQ)

Neuropsychological Symptoms Checklists

Behavioral Observations:

Because traffic resulted in some delay in his commute to the assessment, the patient arrived a few minutes late for both sessions which comprised his evaluation. He was cordial and engaging, and soon established positive rapport with the examiner. He was forthcoming toward all the examiner's questions and requests, and made every effort to be cooperative with test instructions. His speech was normal in tone and in volume. No obvious gross motor difficulties were seen, nor was there any obvious difficulty orienting to his surroundings.

A set of criteria which suggest probable malingering of neurocognitive deficits have gained wide acceptance in the professional community (Slick et al.). They consist of: the presence of a substantial external incentive; two or more types of evidence from neuropsychological testing, excluding definite negative response bias, or, one type of evidence from neuropsychological testing, excluding definite negative response bias; and one or more types of evidence from Self-Report. Also required is that the behaviors meeting these criteria are not fully accounted for by psychiatric, neurological, or developmental factors. (Slick, Sherman, and Iverson, 1999).

Both Symptom Validity tests and the "pattern of performance" method were used to collect data concerning the patient's motivation and consistency of effort. These concern the issues of "performance validity" or "symptom validity". One of these tests found that his ability to recognize words he had just heard was appropriately stronger than the number of words he could remember and report from memory alone (recognition versus free recall).

Data from the Digit Span sub-test also can provide information about the validity of the patient's test performances. The Reliable Digit Span (RDS) result and the total age-corrected scale score (ACSS) were well within the acceptable range (Mathias, C.W. et al., Assessment, 2002, Vol 9:3; Spencer, R.J et al., TCN, 2013, Vol 27:8); both measures indicate a valid response set with no indication of less than optimal effort.

The TOVA Test's Symptom Exaggeration Index (SEI) provides more data concerning effort. On both occasions in which he completed the TOVA, with and without medication, TOVA data show no evidence that he attempted to feign extra difficulty or otherwise magnify his symptoms; the

score indicated “No evidence of Symptom Exaggeration”. This information adds further support to the impression that the patient gave his best effort throughout the evaluation.

The Test of Memory Malingering (TOMM) is intended to discern whether a patient’s effort is reliable and when intentional symptom distortion or exaggeration may have occurred. The patient had a strong performance on the first trial (44/50), and on the second trial he had no errors. These results are very much like those he achieved about two years ago, and do not suggest a lack of effort, nor any attempt to exaggerate symptoms. The TOMM results demonstrate a reliable effort on the patient's part to do his best, which supports this examiner's observations of strong effort and persistence while administering the other tests.

Last, several of the patient's comments indicated that his motivation was strong throughout the evaluation. In summary, there was no indication of any deliberate interference with his best effort or performance during the evaluation, and effort and motivation are considered to have been strong and consistent.

These results are considered a valid indication of his current level of neuropsychological functioning under *optimal* conditions (three sessions of testing, breaks as needed or requested, direct support and encouragement as permitted, few distractions). Everyday functioning would typically be somewhat less efficient due to factors such as fatigue, distraction and frustration.

TEST RESULTS AND INTERPRETATION:

Every effort was made to compare the patient’s performances with similar peers of the same gender, or with persons of similar age, educational level, etc. To that end, a variety of data sources were used, such as A Compendium of Neuropsychological Tests, A Handbook of Normative Data for Neuropsychological Tests, as well as the Halstead-Reitan Neuropsychological Deficit Scale data (NDS, Reitan and Wolfson).

The demographically-corrected “Heaton norms”, (Comprehensive Norms for an Expanded Halstead-Reitan Battery, Heaton et al.) would compare this patient’s performances with males of similar age and level of education. However, some research has called this approach into question. The research raised concerns about the validity and appropriateness of the use of demographically-corrected norms when there has been neurological damage. “Recent research results have shown that when brain damage is present, neuropsychological tests appear to be more affected by brain damage (which they were designed and validated to measure) than to age and education.” (Reitan & Wolfson, 2004). The same investigators also found that the use of such norms can produce seemingly normal scores for individuals who actually have cognitive deficits.

In recognition of this issue, any results derived from demographically-corrected norms will be interpreted with appropriate caution.

A score at the 50th percentile is exactly average, indicating that half the population does better and the other half does worse. There is a broad range of scores around the 50th percentile which falls within the Average range.

Fundamental or Basic Neuropsychological Processes:

Physical - Sensory-Motor:

The performances on the Halstead-Reitan Sensory-Motor tests were largely within normal limits, including tests involving simultaneous stimulation trials. Similarly, his performance on Tactile Finger Perception was at expected levels and fell within normal limits (WNL). However, his Finger Tip Number Writing performances resulted in appreciably more errors than are expected with typical functioning. Also, almost twice as many errors were made with the right hand (11) compared with the left (5). This could suggest a degree of relative inefficiency of the right parietal lobe.

Moving from sensory (input) to motor (output) data, tests of motor speed and strength were WNL overall, compared to other young men in their 20's. Motor speed was above average bilaterally (93rd percentile). Grip strength was in the average range, although this test also showed some mild inefficiency of the right hand (left hemisphere) compared with the left. Compared to other young-adult males with more than high school education, his motor dexterity performances were below the 9th percentile bilaterally, placing those performances in the impaired range.

His total time to complete the Tactile Form Recognition test was unimpaired, by NDS criteria. Also, he made no errors in identifying the various shapes by touch alone, spontaneously and quickly correcting two prospective errors on his left hand.

Cognitive – Attention-Concentration, Immediate Memory and Recapitulation

On the two relevant Halstead-Reitan tests, auditory attention seemed to be without problems. These tests require sustained auditory attention and concentration, and careful discrimination among auditory stimuli. He was asked to identify whether pairs of rhythmic beats were either identical or different (Seashore Rhythm Test), or listen to “nonsense” words and identify the correct written spelling (Speech Sounds Perception Test). NDS norms indicate “no deficit” or “perfectly normal”.

The WAIS Digit Span test involves two different cognitive processes; the repetition span involves immediate attention, while the reversed recitation span involves Working Memory. His overall performance on this test was at the 63rd percentile, which is an improvement from the 37th percentile score he had achieved in 2016. His performance on the repetition span (Digits Forward) portion was low-average.

On two other tests from the Halstead-Reitan Neuropsychological Battery also described as assessing immediate memory and recapitulation, his NDS ratings ranged from “perfectly normal” to mildly impaired. On a simple test of immediate visual attention and memory (Rey 15 Item) his performance was well within expectations. All of these test results are largely WNL, with only a few suggesting some mild difficulty.

The Test of Variables of Attention (TOVA) is an objective computer-based measure of attention performed under carefully calibrated conditions, and compares an individual's results with others of the same age. The patient maintained an adequate and valid response set throughout the 20-minute test, and validity checks within the test indicate that all four quarters of this test are valid.

The TOVA test reports Omission errors, which suggest Inattention, and Commission errors, which suggest Impulsivity. Response Time and other factors are also considered. All TOVA data ought to be interpreted in the context of the individual's level of intellectual functioning; individuals of average intelligence, without attention problems, are expected to produce average scores, while those of above-average intelligence are expected to produce stronger scores. Performances not meeting those expectations suggest difficulty sustaining attention within typical expectations.

In the past, this patient has been prescribed medication for diagnosed ADHD, but he had discontinued it. The TOVA was administered twice, both with and without pre-treatment with Adderall XR. Without medication, the patient made a single inattentive error; he did make several impulsive errors, many of which were during the low-stimulation portion of the test. A similar pattern was seen during the Connors Continuous Performance Test in 2016. On that test, his performance became “more erratic” during the slower-paced portions of the test, lending support to the impression that the patient has more difficulty in low-stimulation or low-demand situations..

His overall Reaction Time on the 2016 Connors was unusually variable and “atypically slow”. This usually indicates that the patient is taking longer than usual to decide on his response (hit or non-hit). A similar pattern was seen on the first TOVA administration. This resulted in an overall TOVA score of -1.54. Scores below zero are more consistent with a group with diagnosed ADHD than a control group. This score was due largely to Response Time, RT Variability and similar variables, as well as a degree of impulsive responding, rather than frank inattentive errors.

As noted, the patient had taken the discontinued ADHD medication prior to the second TOVA administration. In this condition, the patient again made a single inattentive error; he also greatly reduced the number of impulsive errors. Further, the issues with Response Time and RT Variability were not present. The overall result was an ACS score of +2.05.

The data show that the patient is much more prone to respond impulsively in low-stimulation conditions. He is likely to experience these as “boring”, but this examiner’s impression is that he struggles to effectively inhibit “inappropriate” responses in such conditions due to cognitive factors. Also, the IVA+ Plus data from 2014 indicate that the patient had some difficulty with attention, directly, when faced with auditory versus visual material. Based on the 2016 and current data, as well as the information from the IVA+ Plus data, this examiner’s diagnostic impression is ADHD, Predominantly Hyperactive Presentation, Mild. By history this problem was observed from early childhood and pre-dates his concussion history; therefore it is not considered to represent residual effects of any concussion.

Overall, ADHD is an area of some concern for the patient, largely in regard to impulsive responding. He demonstrated an ability to benefit from stimulant medication, as shown by the TOVA data. Without it he had a degree of difficulty that this examiner interprets as Mild, while the medication was effective in mitigating that effect on his cognitive functioning. As always, decisions regarding medication is left to the patient and his consulting or primary care physician.

Visual-spatial and visual-constructional skills:

During the Rey-Osterrieth Complex Figure Test (ROCF), the patient is first asked to copy a “complex figure” carefully and accurately. In a typical administration, a short time later he is asked to draw it again, from memory (brief delay condition); after a longer delay he is asked to draw it a final time (recall condition).

The Copying task provides an indication of visual-perceptual, visual-spatial and fine motor skills. Composing the drawing also calls upon other, more complex skills, such as planning and organization. Some individuals draw from one element, then the neighboring one, while others employ various different approaches. Observing the patient’s approach gives information about the problem-solving skills he spontaneously generates and uses. All these are Executive Functioning [EF] skills, which are discussed later in this report (Higher-Order Cognitive Functions). The later drawing the figure from memory is a test of visual memory, which is discussed later in this report alongside other Memory skills.

His initial Copy of the ROCF was at the 42nd percentile for his age group. This is an improvement over his performance in 2016, when he omitted two areas of detail in the low portion of the drawing; those omissions resulted in a performance in the impaired range. That examiner noted that when the patient was shown the figure and his drawing, he quickly picked up his omissions. To this examiner, that 2016 performance is better understood as reflecting impulsivity, rather than a visual-perceptual problem.

The Beery Visual Motor Integration (VMI) task asks the patient to copy a series of much simpler line drawings than the ROCF drawing. His performance was in the average range at the 30th percentile.

The Hooper Visual Organization Test (VOT) asks the patient to look at a pieces of a series of objects that have been “cut up”, like puzzle pieces (a table, a saw). The patient was to look at them and report what the object would be if the pieces were put together. Therefore this test involves spatial manipulation of mental constructs, reliant on right hemisphere functioning. The patient’s efforts included many more errors than this examiner generally sees, resulting in a performance in the range of Moderate Impairment. The test authors describe this as consistent with “organic brain deficit”, while this examiner’s opinion is that the test is more focused than that general conclusion. Of course, the results are also predicated on adequate focus and attention.

The Rey Tangled Lines Test is a visual scanning test; both speed and accuracy are needed on this test. An error-free performance with an average time of 4-5 seconds per item is typical. The patient’s average time on a sample of the items was 5 seconds, with no errors. This is a WNL performance.

Intermediate Neuropsychological Processes:

Selected Language Skills:

Overall language skills, as they are assessed on the WAIS-III, include learned skills (Vocabulary) and verbal reasoning and problem-solving skills (Similarities). Similarities involves finding a common conceptual element shared by two words (red-blue; lion-bear). The patient’s Similarities

performance was at the 99th percentile; in 2016 it had been at the 84th percentile. His current Vocabulary score is at the 98th percentile; two years ago it had been at the 99th percentile.

It is more likely than not that the differences between the two evaluations do not represent “real” change, but instead are the result of expected variability between administrations of standardized tests (or sub-tests). Overall IQ scores are quite robust and generally tend to be stable over time, but individual sub-tests within the WAIS are more subject to fluctuations from one administration to another. In both evaluations, the patient’s overall verbal skills are an area of relative strength and fall in the Very Superior range.

The patient did not mention any difficulties with word retrieval or word-finding. Results of the Boston Naming Test in 2016 were in the average range in 2016 and were not repeated here.

Memory Performances and Memory Skills:

During the WMS Verbal Paired Associates test, the patient is asked to listen to pairs of words (Gold-Walk, etc.) and learn which words were paired together. The list is repeated four times; learning typically increases over the four trials. The patient’s performance on the first trial was not strong, but that is often seen on this test. He showed a robust learning curve over the four trials, resulting in a score at the

The WMS-III Logical Memory Subtest involves recall of two brief, orally presented stories. Based on how many details he could recall immediately after hearing the stories, the patient’s initial learning and memory was in the average range at the 63rd percentile. The evaluation in 2016, using a slightly different form of this test, found a somewhat stronger performance, at or above the 92nd percentile. Note that he recalled only three more story details at that time.

After a delay, his current performance again fell at the 63rd percentile; in 2016 it again was stronger than currently. In 2016 he could still recall all of the details he’d initially reported, while he dropped a small number of them at this time.

The Rey Auditory Verbal Learning Test (RAVLT) involves a different type of verbal learning. It presents a word list orally to be learned and remembered as well as possible. It permits assessment of verbal learning from several additional perspectives, using five trials with a single word list (List A), and other procedures. The patient had also taken this test in the 2016 evaluation.

His overall performance (Total Words) across five trials was average at the 50th percentile. In 2016, he had a modestly stronger overall performance; five more words overall placed his performance at the 75th percentile. As in 2016, his “learning curve” began slowly on the first trial, falling below expectations compared with persons of similar IQ. (The same initial performance was “average” for the comparison group used in the 2016 analysis, probably using age-group norms). As before, the patient showed a strong learning curve and ability to benefit from repetition over the five trials. He was not quite able to retain all of the words on the list on the last two trials, missing one each time. When presented with a visual list in which the words from the list appeared alongside other words, he correctly identified all of the words from the list, without including any not on the list. Therefore his “miss” of one word on the last two trials could reflect some inconsistency in verbal memory functioning, but it equally could be the result

of “overthinking” the task, or interference from some anxiety, or both. In any case, the differences between his current RAVLT performance and that in 2016 were minor.

In general, the results of the verbal memory tests were above average when compared with peers about his age, and about average when compared with others at about his level of intellectual functioning in the Superior to Very Superior range. He also demonstrated a strong capacity to benefit from repetition of information. His memory for conversation-type material, as seen in the stories, was slightly less strong than his ability to learn from a list (word pairs and RAVLT). This suggests that In everyday life, he (or those around him) may find that he is not as strong at recalling details of conversation as his IQ might suggest he would be. When information conveyed in conversation is personally important, the patient might want to use strategies (“What I heard you say was ... - did I miss anything?”) to assure he has not overlooked important aspects of the communication.

The following tests assess visual memory and learning. Verbal and visual memory abilities involve different processes and neural pathways; as a result, differences between visual and verbal memory functioning may occur.

During the previously administered Rey-Osterrieth Complex Figure Test the person is not told to remember to drawing, but after a two-minute delay the individual is asked to draw it again. This is repeated again after a short time. Since the patient is not told he will be asked to draw the complex figure again, this is a test of “incidental” visual memory. (These were not done at the 2016 evaluation.) After a very brief delay, his redrawing of the figure, scored for the number of relevant details that were included, resulted in a score in the high average range (81st - 84th percentile). After about 20 minutes he lost more of the information, so that his third drawing of the figure fell in the average range (58th - 67th percentile).

The WMS-III Visual Reproduction presents five drawings, one at a time, each of which is shown for a few seconds and then taken away. The patient then attempts to recreate each drawing from memory. These begin with simpler drawings and progress to more complex ones. Overall, these are simpler drawings than the Complex Figure described earlier. On the first, short recall trial, his visual memory performance was at the 92nd percentile. After twenty minutes his reproductions, from memory, contained all of the details he had included in his first drawings. Since most people tend to forget some details, his performance relative to similar-age peers went up to the 99th percentile.

These data suggest that his memory for more complex visual information may decay over a fairly short time span, but is well retained for simpler information.

Intellectual Skills and Academic Achievement:

The patient's performance on selected tests from the Wechsler Adult Intelligence Scales – Third Edition (WAIS-III) resulted in a Full Scale IQ of 136, in the Very Superior Range of intellectual functioning. He achieved a Verbal IQ of 133 and a Performance IQ of 132, both in the Very Superior Range.

Like the previously administered WAIS-IV, the WAIS-III is comprised of four summary “Index” scores: the Verbal Comprehension Index (VCI), Perceptual Reasoning Index (PRI), Working

Memory Index (WMI) and Processing Speed Index (PSI). The Verbal Comprehension Index (VCI) measures verbal knowledge and ability to reason with previously learned information. His score of 131 on this portion of the WAIS was in the Very Superior range at the 98th percentile. He had achieved a score of 132 in the 2016 neuropsychological evaluation.

The Perceptual Organization Index (POI) score reflects an individual's reasoning abilities with visual and hands-on tests. They involve perceptual organization skills and problem-solving. His score of 128 in this area of the test is in the Superior range. This represents a modest improvement over his score of 117 in the 2016 evaluation.

The Working Memory Index (WMI) involves short-term auditory memory, specifically the ability to store information and use it within a few seconds. Working Memory (WM) is critical to effectively learning in school and to many everyday tasks at work and home. ADHD is often associated with corresponding deficits in WM. His performance on these skills resulted in a current score of 126 (96th percentile); his previous score was 111, so his current performance is an appreciable and statistically significant improvement over that performance.

The Processing Speed Index (PSI) is an indicator of general cognitive processing speed. His score for this skill was 125, in the Superior range. This is a significant improvement over the score of 97 he had achieved in 2016.

The improvement in these last two Index scores bears consideration. His previous experience with these tests raises one possibility: that they are due to "practice effects". However, this potential interference with a valid performance is generally considered to be inconsequential by a year from administration of the WAIS, and the time lapse between testing is now about two years, so this possibility is unlikely. Another possibility is that his previous performance represents something less than an optimal performance, since he was given all the neuropsychological tests in one session (whereas the current evaluation took place over two sessions). That, too, seems unlikely to this examiner, since that battery did not involve so many tests and so much time as would be likely to create fatigue for the patient. The last and more likely reason for these performances to be stronger than in 2016 is that the patient enjoyed improvement in his processing speed and working memory since the 2016; that is, that the current stronger performances represent additional recovery of function following his 2015 concussion(s). In this examiner's opinion, that hypothesis seems the most likely explanation for these improvements.

In 2016, the significant disparity among these Index scores caused that examiner to reject the Full Scale IQ as the best indication of overall cognitive ability in favor of using the General Ability Index (GAI), which omits the Processing Speed and Working Memory indices from the calculation. That adaptation is not necessary in the current evaluation, since none of the Index scores are at odds with the others and all are quite strong. Therefore the FSIQ is the best indicator of overall intellectual ability, as already stated.

Some strengths were seen among the tests in the WAIS, and comparisons with the 2016 are available. The patient's very strong performance on a test of visual scanning and decision making was a significant improvement over what he produced on 2016; this is the main reason for the gains seen in Processing Speed. Both verbal and non-verbal tests (sub-tests) involving abstract reasoning showed improvement on this evaluation (currently 96th to 99th percentiles).

The Digit Symbol (Coding) test was, again, his most challenging task. On this test, the patient looks at number-and-symbol pairs in a “key” at the top of the page, and then draws the correct symbols in spaces below, where numbers from one through nine appear in random order. As a paper-and-pencil test that requires drawing symbols efficiently, it is the only WAIS test that bears a similarity to producing manuscript. It is helpful if the person is able to draw the symbols quickly, and it is of further help if the person begins to learn and remember the symbols that go with the numbers. It is evident from the drawn symbols that reproducing them was rather effortful for the patient; this suggests that fine-motor skills are less well-developed, and that producing manuscript most likely is relatively slow and perhaps frustrating for this patient. If he has not already adopted typing as a better alternative for writing, he is encouraged to develop and use typing skills as an alternative.

According to the results from the Academic Achievement Battery, the patient's current level of academic achievement is at the 88th percentile for Reading, at the 75th percentile for Spelling, and at the 96th percentile for written Arithmetic/ Math Calculations. Reading on the AAB measures phonetic reading skills, or the ability to pronounce the word correctly, but not comprehension, knowledge of word meanings or other skills generally thought of as pertinent to reading.

The ABLE Reading Comprehension is an “open book” type of reading comprehension test. The patient's score was at the 88th percentile. The language skills here, as measured by these screening tests, are somewhat below expectations for an individual with his Verbal IQ.

HIGHER ORDER NEUROPSYCHOLOGICAL SKILLS:
Abstract Reasoning, Concept Formation, Problem solving

The skills involved in developing ideas and plans (concept formation), abstract reasoning, drawing inferences and solving novel problems fall at a higher order of neuropsychological functioning. Also at this level are the skills referred to as Executive Functions (EF); these include goal-setting, planning, initiation, self-monitoring and monitoring of progress toward a goal, flexibility, as well as social reasoning and judgment. These skills and abilities are heavily reliant on efficient functioning of frontal and pre-frontal cortex.

The patient was asked to create four oral word lists using two tests, FAS and Word Associations (Animals). He had a starting point or broad category to begin each list. Language skills are involved, but these tests also require initiation, flexibility, rapid processing, and avoiding getting “stuck” by re-starting or re-initiating with new effort. These are Executive Functioning skills, and the frontal lobes are responsible for directing these sorts of behaviors. These tests are related to Executive Functioning and are sensitive to frontal lobe efficiency.

Phonemic fluency (FAS) was at the 80th percentile, in the range, and the categorical fluency performance (Animals) was at the 65th percentile, compared to males in his age group. These results do not suggest that the EF skills involved in them are impaired. Tests were also given in 2016; at that time he was stronger in the categorical fluency test than presently (99th percentile), and less strong (52nd percentile) on the phonemic fluency test.

Trails B is a test of mental flexibility and complex attention; this involves serial shifting of mental set between numbers and letters. These are also EF skills. His performance here was at the 64th percentile. The patient had encountered some difficulty with his visual search on Trails A,

resulting in a relatively slow, 9th percentile performance; however, he had achieved a 75th percentile performance in 2016. While a comparison between Trails A and B can be used to obtain another reference point for EF skills, the current Trails A performance appears to be aberrant, so that comparison was not generated for this interpretation.

The Category Test is a complex test of new problem-solving, judgment, abstract reasoning, mental flexibility and mental efficiency. It is a sensitive measure of abstraction abilities, and requires relatively efficient involvement of a number of 'higher-order' functions. It is also a test of learning ability utilizing non-verbal material. "It is presumably sensitive to cerebral lesions regardless of their location" (Compendium, p.32); some have proposed that in addition it also has relatively greater sensitivity for some skills associated with frontal functioning.

The patient made 32 errors on the test, an unimpaired performance by NDS standards. There would have to be 46 errors or more to rate as impaired by NDS criteria. The demographically-corrected norms would place this performance in the impaired range; however, as noted previously, research has resulted in cautious application of those norms in current practice.

The Tactual Performance Test (TPT) is a complex problem-solving task which requires being blindfolded, so that no visual input is available. The patient places wooden blocks of various shapes into matching spaces within an upright board (tactile-perceptual-motor problem-solving and concept-formation.) There are three trials, first using the dominant hand, then the non-dominant hand, and finally both together. No verbal responses are needed, nor is there any direct visual input; instead, it involves the manipulation of blocks, and the patient relies on their sense of touch.

The patient first uses their dominant hand, then their non-dominant hand, and finally both together. This three-step approach allows the examiner to compare problem-solving skills from each side of the brain by observing separate performances by the left hand, the right hand and both hands together. It is expected that after the patient gains experience with the process on the first trial, the second trial will go more quickly, and improve still more on the third trial.

The patient had considerable difficulty with this task using his right hand on the first trial, compared to others in his age group and IQ level (below 1st percentile). On the next trial, using his left hand, he improved the time to complete the task by 26%, but this was still an impaired performance (below 1st percentile). On the last trial, using both hands, he improved his time by 50%; that performance fell at the 10th percentile, just above the cut-off for an impaired performance.

He attempted to take a thoughtful approach to the task, but when his plan was not effective he didn't adapt by adopting a different approach. This suggests some difficulty with mental/cognitive flexibility when faced with solving a new and novel problem, a difficulty that was also seen in 2016 on a different test of this type. This was a frustrating experience for him, as indicated by sighs and other signs of frustration. Still, he maintained a strong effort and was persistent throughout these procedures, and he did not appear to have any interference from negative emotional reactions, aside from the muted signs already described. This observation suggests that his self-monitoring and emotional regulation (aspects of EF skills) were effective on this task, even as his flexibility was challenged.

After putting the materials away (he never directly saw the board or blocks), he was asked to draw the board and the blocks within it from memory, also drawing the blocks in their proper locations on the board as well as he could remember them. The number of block shapes he drew correctly resulted in a score at the 80th percentile, an unimpaired range by NDS norms. His ability to place or locate them accurately in his drawing was mildly impaired. These data indicate that he remembered the shape of the blocks well, but had more difficulty with their organization on the board.

His performances on this test give the impression of a relative weakness in the EF area of mental flexibility, while suggesting relative strengths in self-monitoring and self-regulation.

Given the number of tests administered, some test performances are expected to fall outside the average range simply due to chance. However, the data set obtained from these results is quite consistent, so that it seems unlikely that the patterns described in the discussion of all the test performances above merely represent a statistical aberration.

Neuropsychological Profile Summary Indices:

The Neuropsychological Deficit Scale (NDS: Reitan & Wolfson, 1986, 1993) was used in assessing the neuropsychological data set obtained in this evaluation. The NDS uses 45 individual scores and four areas of inference to produce an overall composite score for the Halstead-Reitan Neuropsychological Battery. Comparisons can then be made between an individual's performance on that battery and two groups: a clinical group with known brain injury and a group of ordinary control subjects with no known neurological problems. Since the NDS requires all tests in that battery, many of which were not given in 2016, that evaluation could not generate an NDS score for comparison with the current one.

Research comparing the two groups were used to identify "cut-off" scores which most reliably discriminated between the impaired and normal groups. These may be applied to an individual's obtained results in order to determine whether a given score is more typical of the control group or the group with known brain injury; however, due to individual differences, clinical judgment must be used in the application of these cut-off scores to any given individual's evaluation.

The patient's total NDS score was 24. This falls just below the range consistent with Mild Neuropsychological Impairment, which begins with a score of 26. In the NDS research, 8% of the subjects had scores of 25 or less and were considered to be unimpaired brain-damaged subjects. From this data point, this patient's broad, overall level of neuropsychological functioning is more similar to unimpaired subjects than it is to individuals with Mild Neuropsychological Impairment.

The NDS does not include tests outside of the Halstead-Reitan Neuropsychological Battery, whereas this more comprehensive evaluation involved many additional tests (such as the dedicated memory tests). On most of these additional tests the patient demonstrated average and/or adequate performances outside the range indicating impairment. The data set suggests that this patient has continued to enjoy ongoing recovery of function from the accident of record, as

well as any neuropsychological difficulties which may have resulted from his other documented concussions. His overall NDS score is in the Normal range.

PERSONALITY FUNCTIONING:

The patient completed the Millon Clinical Multi-Axial Inventory, Third Edition (MCMI-III) as part of the current evaluation. The MCMI is a contemporary evaluation of psychological functioning and personality attributes that has strong psychometric properties and research-based application. It is intentionally fairly well aligned with the American Psychiatric Association's Diagnostic and Statistical Manual (DSM), currently in its Fifth Edition (DSM-5).

In the course of completing the MCMI, the patient was asked to identify two areas in which he currently experiences some difficulty. From the several choices he was offered, he selected low self-confidence and loneliness as areas of greater concern to him than the other choices.

A tendency toward avoiding self-disclosure was apparent in his response style on this True-False inventory. This could be due to an unwillingness to divulge matters of a personal nature, problematic or not. Alternative reasons for a response style such as this include broad deficits in introspectiveness or psychological-mindedness, or a characterological evasiveness. The scores that were generated were modified to account for the low self-revealing inclinations seen in his response style, so that the resulting scores are valid and interpretable.

Individuals who obtain similar profiles tend to take a measure of delight in defying and challenging convention. They are likely to be strongly opinionated and may be obstinate when challenged. Presenting a facile, easy sociability and charm, such individuals often seek praise and attention; this posture deflects attention from (and, often, personal awareness of) an underlying tense anxiety over being exposed and shamed. When his usual coping mechanisms are challenged or ineffective, he may revert to various forms of acting-out before regaining his characteristic public equanimity.

Individuals who generate this profile are unlikely to seek personal help or interventions (such as psychotherapy, etc) on their own initiative; if they initiate such a request it is likely to be in response to outside persuasion or pressure. Even so, they may respond well to treatment or psychotherapeutic interventions, in either a brief or an extended formulation. A cognitively-based intervention (such as Cognitive-Behavior Therapy) is likely to be productive. This approach could constructively focus on his tendency to devalue others, to think of them as naive and simpleminded and not to trust their judgments. Focused interpersonal methods, such as group and family therapy, may also be helpful, assisting him in viewing himself in a more realistic light and supporting him in learning the value and skills of sharing and cooperation.

Discussion and Summary:

In February of 2015, the patient sustained a mild-complicated Traumatic Brain Injury (TBI). He also has a history of three other reported concussions, including one about nine months after the accident of record. Since then the current evaluation data, in combination with that obtained in February 2016, suggest that he has enjoyed a considerable degree of recovery of neuropsychological skills and capacities.

While he demonstrated impairment of certain specific skills during this evaluation, his overall level of neuropsychological functioning, as reflected in his Neuropsychological Deficit Score (NDS), is below the threshold indicating Mild Neuropsychological Impairment and falls within the “Normal Range”.

Based on the data from the current evaluation, as well as by previous evaluations, this examiner’s professional opinion is that he meets criteria for a diagnosis of ADHD; this appears to be of the Predominantly Hyperactive Type at this time, although by history he previously had met criteria for Combined Type (both inattention and hyperactivity), and also had not been thought to meet criteria for a diagnosis of ADHD. His positive response to medication in this evaluation, based on his performances on a continuous performance test both with and without medication, adds to the impression of the current diagnostic impression.

An area of relative difficulty was seen in Executive Functions (EF). These are higher-order cognitive skills that involve self-regulation, self-management, self-monitoring and task completion. EF skills “manage” or direct other more specific skills, somewhat like an executive might direct and coordinate the activities of a variety of tasks and workers in a corporate environment. EF specifically involves setting goals (identifying a problem) and developing plans to accomplish them; then starting an activity (initiation), monitoring how things are going, and adapting as obstacles or new problems arise along the way.

The patient’s performances on the Tactual Performance Test were in the impaired range, while his performances on Trails B and the fluency tests were in the unimpaired range. These data suggest that he has relatively defined and circumspect difficulties with some EF skills, particularly those involving mental flexibility, while other EF skill areas currently appear to be adequate or stronger.

EF skills are based on the frontal and pre-frontal lobe functioning (as is attention). EF deficits are commonly seen in cases where head trauma has involved blows to the front or back of the head. EF skills are quite important in achieving academic success, as well as vocational stability and satisfaction.

Review of Test Evidence Associated with Neuropsychological Compromise:

There appear to be problems within both right and left cerebral hemispheres, which include involvement of the frontal, parietal and temporal lobes. Test data supporting the impression of neuropsychological deficits include:

Executive Functioning

three consecutive TPT trials

Memory

WMS-IV story tests – below expectations

Attention-Concentration

TOVA responding under high-stimulus condition

Finger-tip Number Writing

Visual Organization and Visual Perception

Hooper VOT

Dysgraphia

Recommendations:

- 1) With the current ADHD diagnosis (Predominantly Hyperactive Presentation), and Mr. Nimer's positive response to a medication trial, he is encouraged to consider use of appropriate medication if he pursues additional education or other pursuits which require extended concentration and focus, at the discretion of his physician.
- 2) If he initiates ADHD medication at some future time, the TOVA could be repeated to aid in assessing the efficacy of the prescribed medication. An objective approach such as the TOVA has advantages over reliance solely on self-report, observational or anecdotal reports for medication evaluation purposes.
- 3) Many recommendations from the 2014 ADHD evaluation could be helpful adjuncts to support Mr. Nimer's sustained attention, and he is encouraged to explore the recommendations that can be integrated into everyday life and determine which are most helpful to him.
- 4) Since a significant risk factor for concussions is a history of having had one (or more), initiating counseling or psychotherapy is recommended to support Mr. Nimer's adaptation to his ongoing recovery, as well as to help him generate alternative behaviors that may reduce the possibility of future concussions. Group or family counseling would also offer substantive benefits, either as an alternative or in addition to individual work.

Thank you for referring this interesting young man as a patient. If there are any questions concerning the issues addressed in this report, please do not hesitate to contact me.

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